

$$E \quad \Theta \quad \Gamma \Lambda \quad - \Theta \quad (\Pi \quad E \quad)$$

ΟΔΗΓΙΕΣ: N . Σ .

1. E f, g . T $f \circ g$:

$$\begin{aligned} \cdot \quad \{x \in D_f \mid g(x) \in D_g\} & \quad \cdot \quad \{x \in D_g \mid f(x) \in D_f\} \\ \cdot \quad \{x \in D_g \mid g(x) \in D_f\} & \quad \cdot \quad \{x \in D_f \mid f(x) \in D_g\} \end{aligned}$$

2. A $\lim_{x \rightarrow x_0} f(x) = -\infty$, x " " x_0 , :

$$\cdot \quad f(x) > 0 \quad \cdot \quad f(x) < 0 \quad \cdot \quad f(x) = 0 \quad \cdot \quad \lim f(x) \in \mathbb{R}$$

3. A $\lim_{x \rightarrow x_0} f(x) = 0$ $\lim_{x \rightarrow x_0} g(x) = +\infty$, $\lim_{x \rightarrow x_0} [f(x)g(x)]$:

$$\cdot \quad \Pi \quad 0 \quad \cdot \quad \Pi \quad +\infty \quad \cdot \quad \Pi \quad -\infty \quad \cdot \quad M \quad 0 \cdot (\pm\infty)$$

4. T $\lim_{x \rightarrow 0} \frac{x}{x}$:

$$\cdot \quad 0 \quad \cdot \quad +\infty \quad \cdot \quad 1 \quad \cdot \quad \Delta$$

5. Γ 1-1 :

$$\begin{aligned} \cdot \quad E \quad \cdot \quad f(x_1) \neq f(x_2) \implies x_1 \neq x_2. \\ \cdot \quad f(x_1) = f(x_2) \implies x_1 = x_2. \quad \cdot \quad T \quad C_f, C_{f^{-1}} \quad y = x. \end{aligned}$$

6. Σ , $f(x)$:

$$\begin{aligned} \cdot \quad M \quad \cdot \quad O \quad f(\alpha) \quad f(\beta) \quad (\quad). \\ \cdot \quad M \quad \cdot \quad M \quad \cdot \quad M \quad \cdot \end{aligned}$$

7. M $[\alpha, \beta]$:

$$\begin{aligned} \cdot \quad M \quad \cdot \quad E \quad 1-1. \\ \cdot \quad \Pi \quad (M) \quad (m) \quad \cdot \quad \Delta \quad \cdot \end{aligned}$$

8. Γ L'Hospital;

$$\cdot \quad \frac{0}{0} \quad \cdot \quad \frac{+\infty}{+\infty} \quad \cdot \quad \frac{-\infty}{+\infty} \quad \cdot \quad 0 \cdot \infty$$

9. A x_0 , :

$$\begin{aligned} \cdot \quad \Upsilon \quad f'(x_0) = 0. \quad \cdot \quad E \quad x_0. \\ \cdot \quad M \quad \cdot \quad \Pi \quad \cdot \end{aligned}$$

10. O $f'(x_0)$:

$$\begin{aligned} \cdot \quad T \quad \cdot \quad T \quad \cdot \\ \cdot \quad T \quad \lambda \quad C_f \quad x_0. \quad \cdot \quad T \quad f. \end{aligned}$$

11. A x_0 f , f x_0 , :

- $f'(x_0) > 0$ $f'(x_0) < 0$ $f'(x_0) = 0$ Δ
12. Π ΔEN Θ Rolle $[\alpha, \beta]$;
 $\text{H } f$ $[\alpha, \beta]$. $\text{H } f$ $[\alpha, \beta]$.
 $f(\alpha) = f(\beta)$. $\text{H } f$ (α, β) .
13. A $f'(x) = 0$ $\text{KA}\Theta\text{E } x$ Δ , :
 $\text{H } f$ Δ . $\text{H } f$ Δ .
 $\text{H } f$ Δ . $\text{H } f$ $a \neq 0$.
14. A $f''(x) > 0$ $x \in \Delta$, f :
 B , B ,
 Δ Σ .
15. T $M(x_0, f(x_0))$:
 $\text{H } f$ $\text{H } C_f$ x_0 .
 $f'(x_0) = 0$ $f(x_0) = 0$.
16. H $\int_{\alpha}^{\beta} f(x)dx = -\int_{\beta}^{\alpha} f(x)dx$:
 $\text{M } \alpha > \beta$. $\text{M } f(x) \geq 0$.
 $\text{E } f$, α, β . Π .
17. T $\int_{\alpha}^{\beta} c dx$ (c), :
 c $c \cdot x$ $c(\beta - \alpha)$ $\beta - \alpha$
18. H $F(x) = \int_{\alpha}^x f(t)dt$ (f) :
 $F'(x) = 0$ $F'(x) = f(t)$ $F'(x) = f(x)$ $F'(x) = f(x) - f(\alpha)$
19. O :
 $\int_{\alpha}^{\beta} f(x)g'(x)dx = f(\beta)g(\beta) - \int_{\alpha}^{\beta} f'(x)g(x)dx$ $\int_{\alpha}^{\beta} f(x)g'(x)dx = [f(x)g(x)]_{\alpha}^{\beta} - \int_{\alpha}^{\beta} f'(x)g(x)dx$
 $\int_{\alpha}^{\beta} f'(x)g'(x)dx = f(x)g(x)$ $\int_{\alpha}^{\beta} (f(x)g(x))'dx = f(x)g(x)$
20. E $f(x) \geq 0$. T C_f , $x'x$ $x = \alpha, x = \beta$ ($\alpha < \beta$) :
 $E = \int_{\beta}^{\alpha} f(x)dx$ $E = -\int_{\alpha}^{\beta} f(x)dx$ $E = \int_{\alpha}^{\beta} f(x)dx$ $E = |f(\alpha) - f(\beta)|$
21. A $\int_{\alpha}^{\beta} f(x)dx = 0$ f $f(x) \geq 0$, :

- . $f(x) = 0 \quad x \in [\alpha, \beta]$. . $\mathcal{H} f$.
 . $\alpha = 0 \quad \beta = 0$. . $\mathcal{H} f$.

22. T $\int_a^\beta f((x)) \cdot (x) dx$:

- . Π . . $A \quad u = x$.
 . $A \quad u = x$. . A .

23. M $\lim[\dots] = 0$. T :

- . $f(x) \cdot (\lambda x + \mu)$. $f(x) - \lambda x$. $f(x) - \lambda$. $f(x) - (\lambda x + \mu)$

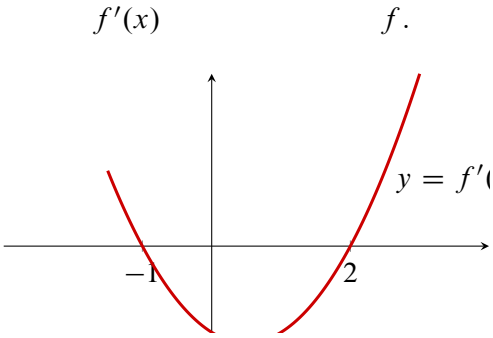
24. H x_0 :

- . O . . $\mathcal{H}$.
 . $T \quad +\infty \quad -\infty$. . O .

25. H $C_f \quad A(x_0, f(x_0)) \quad x'x \quad \omega$. I :

- . $f'(x_0) = \omega$. $f'(x_0) = \omega$. $f'(x_0) = \omega$. $f'(x_0) = \omega$

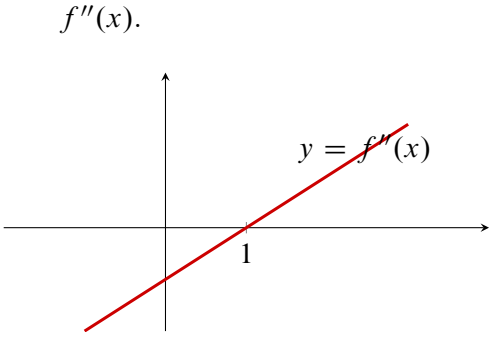
26. II



Σ

$f(x)$;
 $[-1, 2]$ $(-\infty, -1]$ $[2, +\infty)$ $\mathbb{R}$

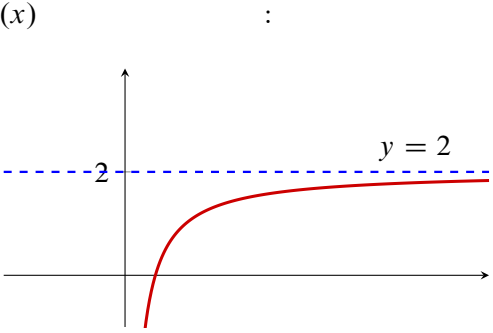
27. Σ



II

$f(x)$ (1);
 H f $x_0 = 1$. T $x_0 = 1$ f .
 H f () $[1, +\infty)$. H

28. Δ

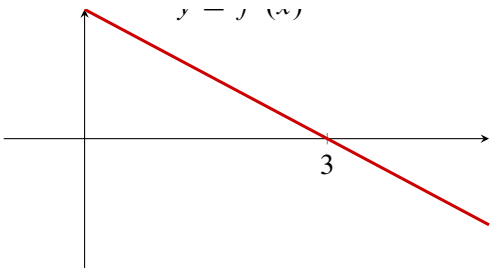


A

, $\lim_{x \rightarrow +\infty} f(x)$;
 $-\infty$ 0 2 $+\infty$

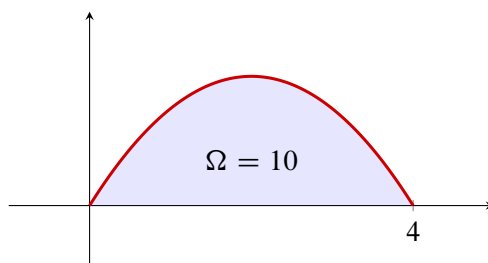
29. A

$f'(x)$, , $x = 3$.



T f $x = 3$;
 . H f f' . . H f f'
 + $-$.
 . E . Δ f' .

30. Δ Ω f , $x'x$ $x = 0, x = 4$.



B ($f(x) > 0$ $(0, 4)$), F f , :
 . $F(4) - F(0) = -10$. $F'(4) - F'(0) = 10$
 . $\int_0^4 f(x)dx = 10 \implies F(4) - F(0) = 10$. $\int_4^0 f(x)dx = 10$

31. A $f'(x_0) = 0$ $f''(x_0) > 0$, x_0 f :
 . Π . Π .
 . Π . Δ .

32. H C_f $(x_0, f(x_0))$:
 . $y = f(x_0) + f'(x)(x - x_0)$. $y - f(x_0) = f'(x_0)(x - x_0)$
 . $y = f'(x_0) \cdot x + f(x_0)$. $y = f(x_0)(x - x_0)$

33. A F_1, F_2 () f Δ , :
 . $F_1(x) = F_2(x)$ $x \in \Delta$. . $F_1(x) - F_2(x) = c$ () $x \in \Delta$.
 . $F_1'(x) \neq F_2'(x)$ x . . $F_1 \cdot F_2 = c$ () .

34. A f $[\alpha, \beta]$ $m \leq f(x) \leq M$ $x \in [\alpha, \beta]$, :
 . $\int_\alpha^\beta f(x)dx = m(\beta - \alpha)$. $m(\beta - \alpha) \leq \int_\alpha^\beta f(x)dx \leq M(\beta - \alpha)$
 . $\int_\alpha^\beta f(x)dx \geq M(\beta - \alpha)$. Δ .

35. H $f(g(x))$:
 . $f'(x) \cdot g'(x)$. $f'(g(x)) \cdot g'(x)$
 . $f(g'(x))$. $f'(g(x)) + g'(x)$

36. To $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$:
 . 0 . 1 . e . $+\infty$

37. Π $f(x) = \ln|g(x)|, \quad g(x) \neq 0 \quad g$;
 . $\frac{1}{g(x)}$. $\frac{g'(x)}{g(x)}$. $\frac{1}{|g(x)|}$. $g'(x) \cdot \ln|g(x)|$
38. A f $[\alpha, \beta] \quad \int_{\alpha}^{\beta} f(x)dx > 0$ ($\alpha < \beta$), :
 . $f(x) > 0 \quad x \in [\alpha, \beta]$. . $\Upsilon \quad x_0 \in [\alpha, \beta] \quad f(x_0) > 0$.
 . $f(\alpha) > 0 \quad f(\beta) > 0$. . $\text{H } f$.
39. A $\lim_{x \rightarrow x_0} f(x) = \ell \quad \lim_{x \rightarrow x_0} g(x) = m$, ΔEN ;
 . $\lim(f+g) = \ell + m$. $\lim(f \cdot g) = \ell \cdot m$
 . $\lim(f/g) = \ell/m$. $\lim(f-g) = \ell - m$
40. A f $[-\alpha, \alpha] \quad (\alpha > 0)$, :
 . $\int_{-\alpha}^{\alpha} f(x)dx = 0$. $\int_{-\alpha}^{\alpha} f(x)dx = 2 \int_0^{\alpha} f(x)dx$
 . $\int_{-\alpha}^{\alpha} f(x)dx = \alpha \cdot f(0)$. $\text{H } f$.
41. K $u = g(x)$, :
 . M . . H .
 . M . . T .
42. A f $[a, b] \quad f(a) \cdot f(b) < 0$, :
 . $\text{H } f$. . $\text{H} \quad f(x) = 0 \quad (a, b)$.
 . $\text{H } f$. . $\text{H } f \quad (a, b)$.
43. A $\lim_{x \rightarrow +\infty} \frac{f(x)}{x} = \lambda \quad \lim_{x \rightarrow +\infty} (f(x) - \lambda x) = \mu, \quad C_f$:
 . K $x = \lambda$. . O $y = \mu$.
 . $\Pi \quad y = \lambda x + \mu \quad +\infty$. . K .
44. A $f'(x_0) \Delta \text{EN} \quad (f \quad x_0), \quad x_0$:
 . $\text{H } f$. . $\text{H } f \quad (\dots f(x) = |x| \quad 0)$.
 . $\text{H } f \quad x_0$. . $\text{H } f \quad x_0$.
45. Π ;
 . $\text{T } \Theta \text{.M.T.} \quad \text{Rolle}, \quad \Theta \text{.M.T.} \quad \text{T Rolle} \quad \Theta \text{.M.T.}, \quad \text{Rolle}$
 $\subset \text{Rolle}$. $\subset \Theta \text{.M.T.}$
 . T . . M Rolle .
46. A f $[\alpha, \beta]$, f :
 . $(f(\alpha), f(\beta))$. $[f(\alpha), f(\beta)]$. $\{f(\alpha), f(\beta)\}$. $\mathbb{R}$

47. To $\int_1^e \frac{1}{x} dx$:

. $e - 1$. 1 . $\frac{1}{e}$. e

48. A f $[-\alpha, \alpha]$ ($\alpha > 0$), $\int_{-\alpha}^{\alpha} f(x) dx$:

. $2 \int_0^{\alpha} f(x) dx$. 0 . $f(\alpha) - f(-\alpha)$. α

49. E f $f'(x) > 0$ x Δ . Π Δ EN ;

. H f Δ . . H f Δ .

. H f Δ . . H f Δ .

50. A $\int_{\alpha}^{\beta} f(x) dx = \int_{\alpha}^{\gamma} f(x) dx + \int_{\gamma}^{\beta} f(x) dx$ (), :

. M $\alpha < \gamma < \beta$. . M $f(x) \geq 0$.

. Γ γ , f . . M f .

ΑΠΑΝΤΗΤΙΚΟ ΦΥΛΛΟ

Σ	Γ	(Κ)
– 1. γ	– 19. β	– 35. β (K)
– 2. β	– 20. γ	– 36. β (B)
– 3. δ	– 21. α	– 37. β (Π -)
– 4. γ	– 22. β	– 38. β (Υ x_0 $f(x_0) > 0$)
– 5. γ	– 23. δ	– 39. γ (A $m = 0$)
– 6. β	– 24. γ	– 40. β (I)
– 7. γ	– 25. β	– 41. β (A)
– 8. δ	– 26. α (H $f' < 0$)	– 42. β (Bolzano)
– 9. β	– 27. β (f'')	– 43. γ (Π)
– 10. γ	– 28. γ	– 44. β (A)
– 11. γ	– 29. β (H f' + -)	– 45. β (Rolle $\subset$ Θ .M.T.)
– 12. β	– 30. γ (T 10)	– 46. β (K , Θ .E.T.)
– 13. β	– 31. β (T f'') ,	– 47. β ($\ln e - \ln 1 = 1$)
– 14. β	– 32. β (T)	– 48. β (I)
– 15. β	– 33. β (Δ)	– 49. γ ($f' > 0$)
– 16. γ	– 34. β (Φ)	– 50. γ (I γ)